CABLE ALGEBRA ACTION AND RANK-ONE FREENESS OF UNKNOT CABLES

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Editorial scope. Three source index slips in the supporting torus-link calculation are preserved verbatim. In original source line 1874 the product index is inconsistent; compare the explicit multi-index convention at 1938. Line 1941 repeats the h-value with qm+r; the author has already derived qm+2r at 1909 and 1931 and the increment q+1 at 1937. At 1985 the printed V_(b-1) denotes the module generator written Y_(b-1) in the preceding equality. These readings point to formulas already present in the author proof; no mathematical formula in this package has been edited.

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Allowed standard prerequisites (editorial).

- BLZ Lemma7.1 and GH Theorem4.3. L-space h-function presentation and cable h formula; retained actual citation and bibliography
- GH and AGL; prop:gradingshifts851–860. L-space property and injectivity of negative-definite cobordisms, and precise grading shifts

0.1. **Heegaard Floer homology.** In this subsection, we review the necessary background for link Floer homology. We assume some familiarity with the basic Heegaard Floer homology and its refinement to knots, see [28, 27, 30, 31] for details. Given a link $\mathcal{L}$ with n components in a 3-manifold Y, one can define the "full" version of Heegaard Floer complex $\mathcal{CFL}(Y,L)$ from the Heegaard diagram by using some version of Lagrangian Floer complexes. The multi-pointed Heegaard link diagram $(\Sigma,\boldsymbol{\alpha},\boldsymbol{\beta},\mathbf{w},\mathbf{z})$ for such pair (Y,L) is defined as follows:

(1) Σ is a closed oriented genus g surface;
(2) $\boldsymbol{\alpha}=\{\alpha_1,\cdots,\alpha_{g+n-1}\}$, and $\boldsymbol{\beta}=\{\beta_1,\cdots,\beta_{g+n-1}\}$ are collections of simple closed curves on Σ such that $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ each span a g-dimensional subspace of $H_1(\Sigma,\mathbb{Z})$ and curves in $\boldsymbol{\alpha}$ (and $\boldsymbol{\beta}$) are pairwise disjoint.
(3) $\mathbf{w}=\{w_1,\cdots,w_n\}$ and $\mathbf{z}=\{z_1,\cdots,z_n\}$ denote sequences of basepoints on Σ such that each component of $\Sigma\setminus\boldsymbol{\alpha}$ (respectively of $\Sigma\setminus\boldsymbol{\beta}$) contains a single point of $\mathbf{w}$ and a single point of $\mathbf{z}$.

The generators of the link Floer chain complex $\mathcal{CFL}(Y,L)$ are tuples of intersection points $\mathbf{x}\in\mathbb{T}_\alpha\cap\mathbb{T}_\beta$ of the Lagrangian tori

$$\mathbb{T}_\alpha=\alpha_1\times\cdots\times\alpha_{g+n-1},\quad \mathbb{T}_\beta=\beta_1\times\cdots\times\beta_{g+n-1}$$

in the symmetric product $\mathrm{Sym}^{g+n-1}(\Sigma)$.

We work over the coefficient field $\mathbb{F}=\mathbb{Z}/2\mathbb{Z}$ in the paper. The marked points z_i and w_i are associated with formal variables V_i and U_i, and the chain complex $\mathcal{CFL}(Y,L)$ is a module over the polynomial ring $R=\mathbb{F}[U_1,\cdots,U_n,V_1,\cdots,V_n]$. The differential in $\mathcal{CFL}(Y,L)$ is obtained by counting pseudo-holomorphic disks in $\mathrm{Sym}^{g+n-1}(\Sigma)$ via:

$$\partial\mathbf{x}=\sum_{\mathbf{y}\in\mathbb{T}_\alpha\cap\mathbb{T}_\beta}\sum_{\varphi\in\pi_2(\mathbf{x},\mathbf{y}),\mu(\varphi)=1}(\#\mathcal{M}(\varphi)/\mathbb{R})U_1^{n_{w_1}(\varphi)}\cdots U_n^{n_{w_n}(\varphi)}V_1^{n_{z_1}(\varphi)}\cdots V_n^{n_{z_n}(\varphi)}\mathbf{y}. \tag{1}$$

Here the sum is taken over all holomorphic disks from $\mathbf{x}$ to $\mathbf{y}$ in $\pi_2(\mathbf{x},\mathbf{y})$ representing the homotopy classes of maps φ from the unit disk $\mathbb{D}\subset\mathbb{C}$ to $\mathrm{Sym}^{g+n-1}(\Sigma)$ satisfying some boundary condition. The Maslov index of φ is denoted by $\mu(\varphi)$. For any point $x\in\Sigma\setminus(\boldsymbol{\alpha}\cup\boldsymbol{\beta})$, $n_x(\varphi)$ denotes the intersection number of the divisor $\{x\}\times\mathrm{Sym}^{g+n-2}(\Sigma)\subset\mathrm{Sym}^{g+n-1}(\Sigma)$ with $\varphi(\mathbb{D})$. The moduli space $\mathcal{M}$ consists of all

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pseudo-holomorphic curves representing φ for a generic 1-parameter family of almost complex structures on $\mathrm{Sym}^{g+n-1}(\Sigma)$. For more details, see [28].

We denote the homology of the link Floer chain complex $\mathcal{CFL}(Y,L)$ by $\mathcal{HFL}(Y,L)$, which is an invariant of the pair (Y,L). The actions of U_iV_i on the complex $\mathcal{CFL}(L)$ are pairwise homotopic, so we introduce the following relation:

$$U_1V_1 = \cdots = U_nV_n = \mathbf{U},$$

and let R_{UV} denote the ring generated by $U_1,\cdots,U_n,V_1,\cdots,V_n$ modulo this relation. Then the action of R on $\mathcal{HFL}(Y,L)$ factors through R_{UV}. We get different versions of Heegaard Floer homology by setting extra constraints on the variables U_i,V_i. For example, $H_*(\mathcal{CFL}(Y,L)/(V_1-1,\cdots,V_n-1))$ is isomorphic to $HF^-(Y)$, the minus version of Heegaard Floer homology of the 3-manifold Y.

In this paper, we focus on links $L=L_1\cup\ldots\cup L_n$ in the three-sphere and, for simplicity, we write the complex as $\mathcal{CFL}(L)$. The complex $\mathcal{CFL}(L)$ and its homology $\mathcal{HFL}(L)$ have several gradings. Firstly, there is the $\mathbb{Q}\times\mathbb{Q}$-valued Maslov bigrading, denoted by $(\mathrm{gr}_{\mathbf{w}},\mathrm{gr}_{\mathbf{z}})$, as well as the Alexander multi-grading $A=(A_1,\ldots,A_n)$ valued in the lattice

$$\mathbb{H}_L=\mathbb{Z}^n+\frac{1}{2}(\ell_1,\ldots,\ell_n)$$

where ℓ_i is the linking number of L_i with the rest of the components, i.e., $\ell_i=\sum_{j\neq i}lk(L_i,L_j)$. These gradings are relatively determined by the following equations:

$$\begin{aligned}
&\mathrm{gr}_{\mathbf{w}}(\mathbf{x})-\mathrm{gr}_{\mathbf{w}}(\mathbf{y})=\mu(\varphi)-2n_{\mathbf{w}}(\varphi)\\
&\mathrm{gr}_{\mathbf{z}}(\mathbf{x})-\mathrm{gr}_{\mathbf{z}}(\mathbf{y})=\mu(\varphi)-2n_{\mathbf{z}}(\varphi)\\
&A_i(\mathbf{x})-A_i(\mathbf{y})=n_{z_i}(\varphi)-n_{w_i}(\varphi)
\end{aligned}\tag{2}$$

where $\varphi\in\pi_2(\mathbf{x},\mathbf{y})$ and $n_{\mathbf{w}}(\varphi)=\sum_{i=1}^n n_{w_i}(\varphi)$ whereas $n_{\mathbf{z}}(\varphi)=\sum_{i=1}^n n_{z_i}(\varphi)$. Furthermore, the actions of U_i,V_i are homogeneous with respect to these gradings with weights

$$(\mathrm{gr}_{\mathbf{w}},\mathrm{gr}_{\mathbf{z}})(U_i)=(-2,0),\quad (\mathrm{gr}_{\mathbf{w}},\mathrm{gr}_{\mathbf{z}})(V_i)=(0,-2),\quad \text{and } A(V_i)=-A(U_i)=\mathbf{e}_i\tag{3}$$

where $\mathbf{e}_i$ is the standard i-th coordinate vector in $\mathbb{R}^n$. The differential on $\mathcal{CFL}(L)$ preserves the Alexander grading, while dropping both $\mathrm{gr}_{\mathbf{w}}$ and $\mathrm{gr}_{\mathbf{z}}$ by one. In fact, these gradings satisfy the relation

$$A_1+\ldots+A_n=\frac{1}{2}(\mathrm{gr}_{\mathbf{w}}-\mathrm{gr}_{\mathbf{z}}).$$

For any $\mathbf{s}\in\mathbb{H}_L$, let $\mathcal{CFL}(L,\mathbf{s})$ denote the subcomplex of $\mathcal{CFL}(L)$ generated by all elements x such that $A(x)=\mathbf{s}$. By the grading formula (3), it is straightforward to see that the product U_iV_i preserves the Alexander gradings, and $\mathcal{CFL}(L,\mathbf{s})$ is a module over the subring $\mathbb{F}[U_1V_1,U_2V_2,\cdots,U_nV_n]$, and so $\mathcal{HFL}(L,\mathbf{s})$ is an $\mathbb{F}[\mathbf{U}]$-module. In particular, for any link L we have

$$\mathcal{HFL}(L)=\oplus_{s\in\mathbb{H}_L}\mathcal{HFL}(L,\mathbf{s}).$$

and $\mathcal{HFL}(L,\mathbf{s})$ is the direct sum of one copy of $\mathbb{F}[\mathbf{U}]$ with some $\mathbf{U}$-torsion.

Recall that a rational homology sphere M is an *L-space* if for each Spin^c-structure $\mathfrak{s}$, $HF^-(M,\mathfrak{s})$ is isomorphic to $\mathbb{F}[\mathbf{U}]$. A link L in the three-sphere S^3 is an *L-space link* if the Dehn surgery $S^3_{\mathbf{d}}(L)$ is an L-space for all $\mathbf{d}\gg 0$. Another way to characterize the L-space links is the condition that $\mathcal{HFL}(L,\mathbf{s})\cong\mathbb{F}[\mathbf{U}]$ for every lattice point $\mathbf{s}\in\mathbb{H}_L$. For any link L in the three sphere, we define the link invariant, known as the h-function.

Definition 0.1. *The h-function $h:\mathbb{H}_L\to\mathbb{Z}$ of a link L in the three-sphere is defined so that for a given $\mathbf{s}\in\mathbb{H}_L$ the value $h(\mathbf{s})$ is $-\frac{1}{2}\mathrm{gr}_{\mathbf{w}}$ for the maximal Maslov grading of non-torsion elements in $\mathcal{HFL}(L,\mathbf{s})$.*

For L-space links L, its h-function can be computed from the Alexander polynomials of the link and its sublinks, see [4]. Moreover, for such links, $\mathcal{HFL}(L)$ is determined by its h-function (see [5]), which is determined by these Alexander polynomials.

Let L be an n-component link in the three-sphere, and $\overline{L}$ be another link obtained from L by adding a full twist on n strands, that is, $\overline{L}$ is obtained from L by blowing down an (-1)-framed unknot circling the n strands of L where we assume that all strands are oriented the same way. Consider the corresponding cobordism from (S^3,L) to $(S^3,\overline{L})$ obtained by attaching a 2-handle to $S^3\times I$ along the aforementioned $(-1)-$framed unknot in $S^3\times\{1\}$. Let $[S^2]$ denote the generator of the second homology class of the cobordism, and let $\phi_k:\mathcal{HFL}(L)\to\mathcal{HFL}(\overline{L})$ denote the induced cobordism maps in homology in the Spin^c-structure $\mathfrak{s}_k$ such that $\langle\mathfrak{s}_k,[S^2]\rangle=2k+1$. See [1] for a detailed discussion of the properties of ϕ_k. We have

Proposition 0.2. [1] *Given an n-component link L in the three sphere, the cobordism maps $\phi_k : \mathcal{HFL}(L) \to \mathcal{HFL}(\overline{L})$ induced from the (-1)-surgery on the unknot with $0 \leq k \leq n-1$ satisfy the following grading properties:*

$$\mathrm{gr}_{\mathbf{w}}(\phi_k) = -k^2 - k,\ A_i(\phi_k) = -k + (n-1)/2,$$

where $i = 1, \cdots, n$.

1. Colored knot Floer homology of the unknot

In this section, we compute the colored knot Floer homology of the unknot.

1.1. **Homology of $T(n, mn)$.** The first step is computing the link Floer homology of the torus link $T(n, nm)$. Since m is fixed in this subsection, we will denote $\mathbf{c}_m = \frac{m(n-1)}{2}$ by $\mathbf{c}$ to simplify notations.

Theorem 1.1. *The homology $\mathcal{HFL}(T(n, mn))$ as a module over R_{UV} is generated by $Y_0, \ldots, Y_{m(n-1)}$ with Alexander and Maslov degrees*

$$A(Y_i) = (\mathbf{c} - i, \ldots, \mathbf{c} - i),\ \mathrm{gr}_{\mathbf{w}}(Y_i) = -(q+1)(qm + 2r), \quad i = qm + r,\ 0 \leq r \leq m-1.$$

Furthermore, the relations between Y_i are spanned by $U_I Y_i = V_{\overline{I}} Y_{i+1}$ for all subsets $I \subset \{1, 2, \cdots, n\}$ with $q+1$ elements. Here, $\overline{I} = \{1, 2, \cdots, n\} \setminus I$, $U_I = \prod_{j \in I} U_i$ and $V_{\overline{I}} = \prod_{j \in \overline{I}} V_j$.

Proof. The proof is similar to the proof of [5, Theorem 7.3]. By [5, Lemma 7.1], to write down the generators we need to compute the h-function for $T(n, mn)$. Since the Alexander polynomial for $T(n, mn)$ is symmetric in $t_1, \ldots, t_n$, the h-function is symmetric in $s_1, s_2, \cdots, s_n$ so we consider the case $s_1 \leq s_2 \leq \cdots \leq s_n$. By [13, Theorem 4.3] for $s_1 \leq s_2 \leq \cdots \leq s_n$ we get

$$h(s_1, \ldots, s_n) = h_0(s_1 - \mathbf{c}) + h_0(s_2 - \mathbf{c} + m) + \ldots + h_0(s_n - \mathbf{c} + m(n-1)) \tag{4}$$

where

$$h_0(x) = \begin{cases} 0 & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

In particular, for $\mathbf{c} \leq s_1 \leq \ldots \leq s_n$ we get $h(s_1, \ldots, s_n) = 0$. Suppose that $s = \mathbf{c} - i$ where $i = mq + r \geq 0$ then (assuming $i \leq m(n-1) = 2\mathbf{c}$) we get

$$s - \mathbf{c} \leq 0, s - \mathbf{c} + m \leq 0, \ldots, s - \mathbf{c} + qm \leq 0 \text{ but } s - \mathbf{c} + (q+1)m > 0$$

so

$$h(s, \ldots, s) = (q+1)i - \frac{1}{2}q(q+1)m = \frac{1}{2}(q+1)(qm + 2r).$$

We claim that

$$h(s-1, \ldots, s-1) - h(s, \ldots, s) = q + 1. \tag{5}$$

If $r < m - 1$ then going from s to $s-1$ changes i to $i+1$ and r to $r+1$ while preserving q, so (5) is clear. If $r = m - 1$ then $i = qm + (m-1), i + 1 = (q+1)m + 0$ and

$$h(s, \ldots, s) = \frac{1}{2}(q+1)(qm + 2m - 2),\ h(s-1, \ldots, s-1) = \frac{1}{2}(q+2)(q+1)m$$

hence (5) holds as well.

Furthermore, one can check that

$$h(\underbrace{s-1, \ldots, s-1}_{k}, s, \ldots, s) = \begin{cases} h(s, \ldots, s) + k & k \leq q+1 \\ h(s-1, \ldots, s-1) = h(s, \ldots, s) + q + 1 & k \geq q+1. \end{cases}$$

Let Y_i be the generator of the $\mathbb{F}[\mathbf{U}]$-tower in Alexander grading $(\mathbf{c} - i, \ldots, \mathbf{c} - i)$. It will have Maslov grading

$$-2h(\mathbf{c} - i, \ldots, \mathbf{c} - i) = -(q+1)(qm + 2r)$$

and by [5, Lemma 7.1], $Y_0, Y_1, \cdots, Y_{m(n-1)}$ generate $\mathcal{HFL}(T(n, mn))$.

By [5, Lemma 7.1], the relations are spanned by the relations between Y_a and Y_b for any $0 \leq a \leq b \leq (n-1)m$ which are the ones spanned by $(U_i V_i + U_j V_j) Y_a$ and $(U_i V_i + U_j V_j) Y_b$ as well as sums

$$U^I Y_a + V^J Y_b$$

ranging over sequences of nonnegative integer $I = (I_1, \cdots, I_n)$ and $J = (J_1, \cdots, J_n)$ such that

$$I + J = (b - a, \cdots, b - a) \text{ and } |I|_{L^1} := \sum_{k=1}^{n} I_k = (q_b + 1)(q_b m + 2r_b)/2 - (q_a + 1)(q_a m + 2r_a)/2$$

where $a = q_a m + r_a$ and $b = q_b m + r_b$. Here, $U^I = \prod_k U_k^{I_k}$ and $V^J = \prod_k V_k^{J_k}$.

If $a = i$ and $b = i+1$, then $(q_b+1)(q_b m + r_b)/2 - (q_a+1)(q_a m + r_a)/2 = q+1$ and so these relations are exactly the ones described in the statement.

Suppose $b > a+1$. We show that the relations between Y_a and Y_b are in the span of the relations between consecutive Y_i and Y_{i+1}. We prove this claim by induction on b. First, we show that there is a tuple $I' = (I'_1, \cdots, I'_n)$ such that for all $1 \le i \le n$, $0 \le I'_i \le I_i$, and

$$|I'|_{L^1} = q_{b-1} + 1, \quad |I'|_{L^\infty} = \max\{I'_1, \cdots, I'_n\} \le 2 \text{ and } |I - I'|_{L_\infty} \le b - a - 1$$

For $k \in \mathbb{N}$, we write

$$a_k := \#\{i : I_i \ge k\}.$$

Then

$$\sum_{k=1}^{b-a} a_k = |I|_{L^1} = (q_b+1)(q_b m + 2r_b)/2 - (q_a+1)(q_a m + 2r_a)/2, \quad \text{and } a_1 \ge a_2 \ge \cdots \ge a_{b-a},$$

and so by Equation (5) $\sum_{k=1}^{b-a} a_k = \sum_{i=a}^{b-1}(q_i+1)$. Consequently, the number of entries I_i that are equal to $b-a$ is at most $q_{b-1}+1$. So, such I' exists if and only if $a_1 + a_2 \ge q_{b-1}+1$. Suppose to the contrary that $a_1 + a_2 < q_{b-1}+1$. Then $a_2 < (q_{b-1}+1)/2$ and therefore

$$\sum_{i=a}^{b-1}(q_i+1) = \sum_{k=1}^{b-a} a_k = a_1 + a_2 + \sum_{k=3}^{b-a} a_k < q_{b-1} + 1 + (b-a-2)\frac{q_{b-1}+1}{2} = (b-a)\frac{q_{b-1}+1}{2}.$$

On the other hand, $q_{b-1}+1 \le q_{b+a-1}+1 \le (q_{a+i}+1) + (q_{b-i-1}+1)$ implies that

$$\sum_{i=a}^{b-1}(q_i+1) \ge \frac{b-a}{2}(q_{b-1}+1)$$

which is a contradiction.

Therefore such a tuple I' exists. By induction, for $J' = (b-a-1, \cdots, b-a-1) - (I - I')$, the relation

$$U^{I-I'} Y_a = V^{J'} Y_{b-1}$$

is in the span of the claimed relations between consecutive generators. Then, this relation implies that

$$U^I Y_a = U^{I'} U^{I-I'} Y_a = U^{I'} V^{J'} Y_{b-1}.$$

If $|I'|_{L^\infty} = 1$, the relations between Y_{b-1} and Y_b imply that

$$U^I Y_a = U^{I'} V^{J'} Y_{b-1} = V^J Y_b.$$

Otherwise, for any index i with $I'_i = 2$, we observe that $(I - I')_i < b - a - 1$, hence $J'_i > 0$. Then $U^{I'} V^{J'}$ contains a factor of $U_i V_i$. Since, $|I'_i|_{L_1} = q_{b-1} + 1 \le n$ there is an index j with $I'_j = 0$. We will trade the $U_i V_i$ factor with $U_j V_j$. By repeating this process enough times, we obtain another tuple I'' from I' such that $|I''|_{L^1} = |I'|_{L^1}$ and $|I''|_{L^\infty} = 1$. That is

$$U^{I'} V^{J'} Y_{b-1} = U^{I''} V^{J''} V_{b-1} = V^J Y_b.$$

□

1.2. **Cable algebra and cobordism maps.** Next, we explicitly describe the cobordism maps ϕ_k corresponding to adding full twists for $T(n, mn)$. In order to do that, we consider the direct sum of homology for all n-strand cables of the unknot with nonnegative integer slopes:

$$\mathrm{Cab}_n(O) := \mathcal{TW}(O_n) = \bigoplus_{m=0}^{\infty} \mathcal{HFL}(T(n, nm)) \tag{6}$$

where O is the unknot and O_n is the n-component unlink.

Each summand is $\mathbb{Z}^n \oplus \mathbb{Z}$ graded by Alexander multi-grading ($\mathbb{Z}^n$) and Maslov grading ($\mathbb{Z}$). Moreover, $\mathrm{Cab}_n(O)$ has an additional $\mathbb{Z}$ grading given by the cabling slope m, we call it *twist grading* and denote it by **tw**. Note that cobordism maps ϕ_k, corresponding to adding a full twist, act naturally on $\mathrm{Cab}_n(O)$ with twist grading 1 i.e. they map $\mathcal{HFL}(T(n, mn))$ to $\mathcal{HFL}(T(n, (m+1)n))$.

Definition 1.2. *The n-strand* **cable algebra** $\mathcal{A}_n$ *is defined as a $\mathbb{Z}^n \oplus \mathbb{Z} \oplus \mathbb{Z}$ graded algebra over R_{UV} with commuting generators $a_0, \ldots, a_{n-1}$ and the following relations:*

- *Linear relations*

$$U_I a_{k-1} = V_{\overline{I}} a_k \tag{7}$$

where $I \subset \{1,\dots,n\}$ is an arbitrary subset with $|I| = k$ and $\overline{I} = \{1,\dots,n\} \setminus I$.

- *Quadratic relations*

$$a_i a_j = \mathbf{U}^{k\ell - ij} a_k a_\ell \tag{8}$$

whenever $i + j = k + \ell$ and $i \le k \le \ell \le j$.

As above, the gradings correspond to the Alexander ($\mathbb{Z}^n$) and Maslov ($\mathbb{Z}$) gradings, along with an additional $\mathbb{Z}$ twist grading. *The generator a_k has Alexander grading*

$$A(a_k) = \left(\frac{n-1}{2} - k, \dots, \frac{n-1}{2} - k\right),$$

Maslov grading $\mathrm{gr}_{\mathbf{w}}(a_k) = -k^2 - k$ and twist grading $\mathbf{tw}(a_k) = 1$.

It is easy to check that the relations are homogeneous with respect to all three gradings.

Theorem 1.3. *The homology $\mathrm{Cab}_n(O)$ is a $\mathbb{Z}^n \oplus \mathbb{Z} \oplus \mathbb{Z}$ graded module over the algebra $\mathcal{A}_n$ where the generators a_k act by cobordism maps ϕ_k. Furthermore, $\mathrm{Cab}_n(O)$ is a free $\mathcal{A}_n$-module generated by a single class $1 \in \mathcal{HFL}(T(n,0))$.*

Proof. We proceed in several steps.

Step 1: We show that there is a well-defined action of algebra $\mathcal{A}_n$ on $\mathrm{Cab}_n(O)$, where a_k acts by the cobordism map ϕ_k. First, we prove that the actions of ϕ_k and $\phi_{k'}$ on $\mathrm{Cab}_n(O)$ commute. By [13] all links $T(n,nm)$ are L-space links, so $\mathcal{HFL}(T(n,mn))$ is free of rank 1 over $\mathbb{F}[\mathbf{U}]$ in each Alexander degree. The maps ϕ_k and $\phi_{k'}$ correspond to negative definite cobordisms, so they are both injective [1], and hence determined by their Alexander and Maslov degrees. Now we observe that both compositions $\phi_k \circ \phi_{k'}$ and $\phi_{k'} \circ \phi_k$ have the same Alexander and Maslov degrees by Proposition 0.2, and hence coincide.

Next, we check that these cobordism maps satisfy relations (7) and (8). Note that a_k has the same Alexander, Maslov and twist grading as ϕ_k, thus relations (7) and (8) being homogeneous, implies that the cobordism maps satisfy these relations as well and the action is well-defined.

Step 2: We prove that $\mathrm{Cab}_n(O)$ is generated by a single class $1 \in \mathcal{HFL}(T(n,0))$ under the action of $\mathcal{A}_n$. We follow Theorem 1.1 and claim that the generator $Y_i \in \mathcal{HFL}(T(n,mn))$ can be identified with the monomial $a_q^{m-r}a_{q+1}^r$ (where $i = mq + r$) applied to 1. By the above, the element $a_q^{m-r}a_{q+1}^r(1)$ is nonzero, and it is sufficient to verify that it has the same Alexander and Maslov degree as Y_i. Indeed, the Alexander grading equals

$$A_j(a_q^{m-r}a_{q+1}^r) = (m-r)\left(\frac{n-1}{2} - q\right) + r\left(\frac{n-1}{2} - q - 1\right) = m\frac{n-1}{2} - mq - r = \mathbf{c} - i$$

and the Maslov grading equals

$$\mathrm{gr}_{\mathbf{w}}(a_q^{m-r}a_{q+1}^r) = -(m-r)q(q+1) - r(q+1)(q+2) = -(q+1)(mq+2r).$$

To sum up, we have a surjective map of $\mathcal{A}_n$ modules from $\mathcal{A}_n$ to $\mathrm{Cab}_n(O)$.

Step 3: Finally, we need to prove that this map is an isomorphism and $\mathrm{Cab}_n(O) \simeq \mathcal{A}_n$. For this, we need to find the generators of $\mathcal{A}_n$ over R_{UV}, relations between them and compare these with Theorem 1.1.

Define $\widetilde{Y}_i = a_q^{m-r}a_{q+1}^r \in \mathcal{A}_n$ where $i = mq + r$. Let us prove that $\widetilde{Y}_i$ generate $\mathcal{A}_n$ over R_{UV}. Indeed, consider a monomial $M = a_{i_1} \cdots a_{i_m}$ with $i_1 \le i_2 \le \cdots \le i_m$. If $i_m - i_1 \le 1$ then $M = \widetilde{Y}_{i_1+\dots+i_m}$ by the above. Otherwise we can use (8) to write

$$a_{i_1}a_{i_m} = \mathbf{U}^{i_m - i_1 - 1}a_{i_1+1}a_{i_m-1}.$$

since $(i_1+1)(i_m-1) - i_1 i_m = i_m - i_1 - 1$. This decreases the power of a_{i_1} and we can proceed by induction.

Finally, we need to verify that the relations from Theorem 1.1 hold for $\widetilde{Y}_i$. This is clear since the relations between $\widetilde{Y}_i = a_q^{m-r}a_{q+1}^r$ and $\widetilde{Y}_{i+1} = a_q^{m-r-1}a_{q+1}^{r+1}$ agree with the relations between a_q and a_{q+1}, up to multiplication by $a_q^{m-r-1}a_{q+1}^r$. □

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